Master's Thesis Exposé

Physics-Informed Multi-Modal Transformer Architectures for Robust Industrial Welding Anomaly Detection

1 Abstract

Industrial production systems increasingly generate heterogeneous data from cameras, microphones, machine sensors, process logs, and inspection devices. The practical use of such data for anomaly detection depends not only on the performance of individual machine learning models, but also on the design of a suitable multimodal AI architecture [1, 2]. In real industrial settings, sensor streams may be temporally misaligned, partially missing, noisy, or degraded. This creates a need for architectures that can combine multiple modalities while maintaining stable performance under such conditions.

This thesis investigates multimodal Transformer-based architectures for robust industrial anomaly detection using the Intel Robotic Welding Multimodal Dataset as a case study. Transformer architectures are relevant because attention mechanisms can support representation learning and fusion across heterogeneous inputs [3]. The Intel dataset provides video, audio, sensor time-series data, and post-weld images from an industrial welding environment [4, 5]. Welding is treated as the industrial application domain, while the thesis focus remains on AI architecture, multimodal data processing, fusion strategy, and robustness evaluation. The thesis does not aim to optimize welding parameters, conduct metallurgical modelling, or replace expert welding assessment.

In this exposé, the term "physics-informed" is used in a restricted sense: it refers to the use of process-related sensor and metadata context within the AI architecture, rather than to explicit physical simulation or equation-based modelling. A short review of suitable Transformer-based representation models will be conducted to justify the final architecture selection. Candidate model families include Vision Transformers for visual representation, Video Swin Transformer and TimeSformer for video representation, PatchTST for time-series representation, Audio Spectrogram Transformer for audio representation, RT-DETR as a possible visual representation candidate, and attention-bottleneck approaches for multimodal fusion [6, 7, 8, 9, 10, 11, 12]. The implementation will focus on a small number of feasible pretrained or lightweight components, with separate branches for visual, audio, and sensor time-series data.

The work compares unimodal baselines, a simple concatenation-fusion baseline, a modular late-fusion architecture, and a process-aware gated cross-modal fusion architecture. The expected outcome is a documented prototype pipeline that shows how different fusion strategies behave on the same multimodal industrial dataset, particularly under missing or degraded modality conditions.

2 Introduction and Motivation

Modern industrial quality monitoring increasingly relies on multimodal data. Visual data can capture process or product appearance, audio can reflect acoustic behaviour, sensor logs describe machine states, and post-weld images provide information about final weld appearance. However, these data sources differ in format, sampling frequency, dimensionality, reliability, and preprocessing requirements. Therefore, one central AI and data-engineering challenge is how such heterogeneous data can be integrated into a reproducible and robust anomaly detection pipeline [1, 13].

Many industrial AI approaches focus mainly on predictive accuracy. In practical settings, however, robustness, modularity, interpretability, and maintainability are also important. A model may perform well when all sensors are available, but fail when one modality is missing, delayed, noisy, or degraded. This is especially relevant in industrial environments where sensors can fail, data streams can be incomplete, and measurement quality can vary.

The Intel Robotic Welding Multimodal Dataset provides a suitable case study because it contains multiple data modalities from an industrial production-related setting [4, 5]. The thesis uses this dataset to evaluate multimodal AI architecture design, not to conduct welding-engineering research. The key motivation is to investigate how multimodal data can be structured, represented, fused, and evaluated under controlled missing-modality and degradation scenarios.

3 Research Questions

Main Research Question

How can a process-aware multimodal AI architecture be designed and evaluated for robust industrial anomaly detection using robotic welding data as a case study?

Sub-Question 1

Which lightweight or pretrained Transformer-based representation components are suitable for visual, audio, and sensor time-series data in a multimodal industrial anomaly detection pipeline?

Sub-Question 2

How do unimodal baselines, simple concatenation fusion, modular late fusion, and process-aware gated cross-modal fusion compare in terms of predictive performance?

Sub-Question 3

Can a gated cross-modal fusion architecture improve robustness when one or more modalities are missing, degraded, or unreliable?

Sub-Question 4

How can multimodal architecture performance be evaluated beyond accuracy, including

robustness, interpretability, modularity, and practical usability?

4 Aim and Objectives

The aim of this thesis is to design, implement, and evaluate a multimodal AI architecture for robust industrial anomaly detection using the Intel Robotic Welding Multimodal Dataset as a case study.

The specific objectives are:

1. To analyse the dataset structure from an AI and data-engineering perspective.
2. To design a modular pipeline for multimodal data ingestion, preprocessing, representation extraction, fusion, and evaluation.
3. To review candidate Transformer-based architectures for visual, audio, sensor, and multimodal fusion tasks.
4. To select feasible lightweight or pretrained components for implementation instead of training large foundation models from scratch.
5. To implement unimodal baselines for visual, audio, and sensor-based anomaly or defect detection.
6. To implement a simple concatenation-fusion baseline.
7. To implement and compare two main fusion architectures: modular late fusion and process-aware gated cross-modal fusion.
8. To compare the proposed fusion architectures against unimodal and concatenation baselines.
9. To evaluate the architectures using predictive metrics such as accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrix.
10. To perform controlled robustness tests by removing or degrading selected modalities.
11. To derive practical design recommendations for multimodal industrial AI systems under sensor uncertainty.

5 Methodology

The methodology follows an AI system and data-architecture development approach. The thesis does not focus on welding physics, welding process optimization, or metallurgical interpretation. Instead, it focuses on multimodal data processing, model architecture selection, fusion strategy comparison, and robustness evaluation.

5.1 Dataset Understanding and Learning Setup

The Intel Robotic Welding Multimodal Dataset will first be analysed to understand its available modalities, file structure, labels, metadata, and sample-level relationships. The dataset contains video, audio, sensor time-series data, and post-weld images [4, 5]. These modalities will be mapped into a common sample-level structure so that each sample can be consistently processed across the pipeline.

The primary learning setup will be formulated as supervised classification using the available quality or defect labels. If class imbalance or label sparsity limits multiclass evaluation, the task will additionally be analysed in a binary good-versus-defective setting or through anomaly scores. This keeps the thesis feasible while still allowing a meaningful evaluation of the proposed architectures.

5.2 Preprocessing and Representation Pipeline

A modular preprocessing pipeline will be implemented. Video data may be sampled into representative frames or short clips. Post-weld images will be resized and normalized. Audio data will be transformed into spectrogram-based representations. Sensor logs will be normalized and segmented as time-series data.

Candidate visual encoders include Vision Transformer, Video Swin Transformer, TimeSformer, and RT-DETR [6, 7, 8, 11]. RT-DETR will therefore only be considered if its pretrained visual backbone can be used meaningfully for representation extraction; it will not be treated as a supervised object detector unless suitable localization annotations are available. The practical implementation will prioritize feasible pretrained visual encoders for image or video representation.

For sensor time-series data, PatchTST will be considered as a suitable Transformer-based encoder because it represents time series through patches and can be adapted for classification or representation learning [9]. For audio, a lightweight spectrogram-based encoder or pretrained audio representation model may be used, with Audio Spectrogram Transformer serving as a relevant reference architecture [10]. The scope will remain focused on reusable representations and fusion architecture, not on training large models from scratch.

5.3 Baseline Models

Before implementing the two main fusion architectures, several baselines will be created. These baselines are important because they allow the proposed fusion methods to be evaluated against simpler alternatives.

The baseline setup will include:

1. **Unimodal visual baseline:** uses only post-weld images or sampled video frames.

2. **Unimodal sensor baseline:** uses only sensor time-series data.
3. **Unimodal audio baseline:** uses only audio-derived spectrogram features.
4. **Simple concatenation-fusion baseline:** concatenates modality embeddings into one feature vector before classification.

These baselines provide a clear reference point for evaluating whether more advanced fusion architectures provide measurable benefits. The main comparison variable is the fusion strategy: unimodal modelling, simple embedding concatenation, modular late fusion, and process-aware gated cross-modal fusion.

5.4 Architecture A: Modular Late Fusion

The first main architecture is a modular late-fusion architecture. In this setup, each modality is processed by its own branch and produces either a prediction score or compact representation. The final output is generated by combining the modality-level predictions at the decision level.

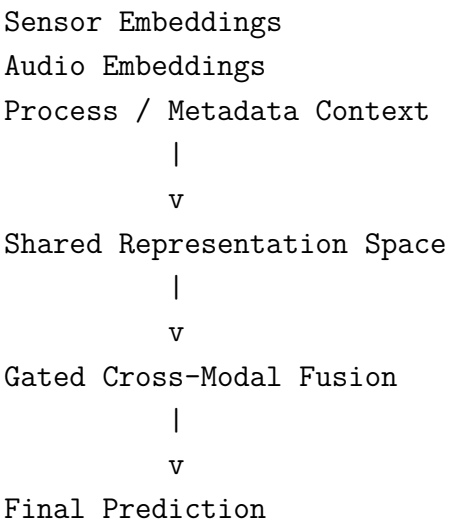
```
Video / Image Branch  -> Visual Prediction
Sensor Branch         -> Sensor Prediction
Audio Branch          -> Audio Prediction
                              |
                              v
                      Late-Fusion Layer
                              |
                              v
                      Final Prediction
```

This architecture is expected to be relatively easy to implement, modular, and interpretable. It allows each modality to be analysed separately and supports missing-modality testing because branches can be removed or disabled.

5.5 Architecture B: Process-Aware Gated Cross-Modal Fusion

The second main architecture is a process-aware gated cross-modal fusion architecture. In this setup, each modality-specific encoder produces an embedding. These embeddings are projected into a shared representation space and fused using a gating mechanism. The gate learns to weight modality contributions depending on the available information and estimated reliability of each modality.

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Visual Embeddings
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The term "process-aware" means that available industrial context, such as sensor-derived features, metadata, or modality availability indicators, can be used to support fusion. It does not mean detailed physical welding simulation or physics-equation-based modelling.

The gated fusion architecture is expected to be more robust than static fusion because it can reduce the influence of missing, noisy, or less reliable modalities. Related multimodal fusion approaches, such as attention bottlenecks, provide relevant architectural motivation for this type of fusion design [12].

5.6 Robustness and Ablation Evaluation

Robustness will be evaluated through controlled modality-ablation and corruption tests. These experiments will simulate realistic uncertainty conditions. Robustness will be tested through controlled ablations, including removal of audio, sensor, or visual inputs, addition of noise to sensor signals, degradation of visual inputs, and simulated temporal shifts between modalities.

The goal is to assess whether the architecture degrades gracefully rather than failing completely when modalities become unreliable.

5.7 Evaluation Metrics

Predictive evaluation will include accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrix. Architecture-level evaluation will include modularity, robustness to missing modalities, interpretability of modality contribution, reproducibility, extensibility, and implementation feasibility.

This ensures that the thesis evaluates not only model performance but also the practical suitability of the architecture.

6 Expected Contribution

The expected contribution of this thesis is a compact and feasible reference architecture for multimodal industrial anomaly detection under sensor uncertainty.

The thesis is expected to deliver:

1. A structured review of suitable Transformer-based components for multimodal industrial data.
2. A reusable multimodal AI pipeline for visual, audio, and sensor data.
3. Unimodal and concatenation-fusion baselines.
4. A comparison of modular late fusion and process-aware gated cross-modal fusion.
5. A robustness evaluation under missing, degraded, and misaligned modality conditions.
6. Practical guidelines for designing multimodal industrial AI systems.
7. A clear separation between AI architecture design and welding-engineering analysis.

Scope limitation: This thesis does not aim to train large foundation models from scratch. The focus is on pipeline design, representation extraction, fusion strategy, robustness evaluation, and architecture comparison using lightweight or pretrained components. The thesis also does not aim to optimize welding parameters, model metallurgical defect causes, or provide expert-level welding interpretation. Welding is used as an industrial case study for evaluating multimodal AI architecture under controlled missing-modality and degradation scenarios.

7 Timeline and Milestones

Table 1: Planned timeline and milestones for the Master's thesis.

Phase	**Milestones / Deliverables**
Month 1: Literature Review and Scope Definition	Review multimodal AI systems, industrial anomaly detection, Transformer-based representation models, fusion strategies, and robustness under missing modalities. Finalize selected components.
Month 2: Dataset and Pipeline Design	Analyse dataset structure, define sample mapping, implement data loading and preprocessing pipeline.
Month 3: Baseline Implementation	Implement unimodal baselines and simple concatenation-fusion baseline.
Month 4: Fusion Architecture Implementation	Implement modular late fusion and process-aware gated cross-modal fusion.
Month 5: Evaluation and Robustness Testing	Evaluate predictive performance, modality ablation, corruption tests, missing-modality robustness, and interpretability.
Month 6: Documentation and Finalization	Document architecture, experiments, results, limitations, and recommendations. Finalize thesis writing.

8 References

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